Xi'an Jiaotong-Liverpool University

西交利物浦大学

PAPER CODE	EXAMINER	DEPARTMENT	TEL
INT305		Intelligent Science	

1st SEMESTER 2024/25 Resit Exam

BACHELOR DEGREE – Year 4

Machine Learning

TIME ALLOWED: 3 Hours 0 Minutes

INSTRUCTIONS TO CANDIDATES

1、 Total marks available are 100.

2、 Answer all questions.

3、 The number in the column on the right indicates the approximate marks for each section.

4、 In general, it is particularly important to give reasons for your answer. Only partial marks will be awarded for correct answers with inadequate reasons.

5、 Answer should be written in the answer booklet(s) provided.

6、 The university approved calculator - Casio FS82ES/83ES can be used.

7、 Only solutions written in English will be accepted.

Answer All Questions

1. The Zero-One Loss is a simple classification loss function that intuitively counts how many mistakes the model makes. In this problem we will examine the Zero-One Loss in the context of binary classification, where the loss for a single example x_i with label $y_i \in \{0,1\}$ is given by:

$$L(x_i) = \mathbb{1}[s_{y_i} - s_{(1-y_i)} < 0]$$

where s_j represents the model's predicted score for the $j-th$ class and is an indicator function that is 1 if the condition inside the function is true, and 0 otherwise.

Now, A friend of yours suggests that you can train a linear model classifier by performing gradient descent over the zero-one loss function. Is this a good idea? Briefly explain why or why not.

Total 12

2. Pooling units take n values $x_i,\ i \in [1, n]$ and compute a scalar output whose value is invariant to permutations of the inputs.

(1) The Lp-pooling module takes positive inputs and computes $y = (\sum_i x_i^p)^{\frac{1}{p}}$, assuming we know that $y' = \frac{\partial L}{\partial y}$, what is $x_i' = \frac{\partial L}{\partial x_i}$? **(8')**

(2) The log-average module computes $y = \frac{1}{\beta} ln(\frac{1}{n} \sum_i exp(\beta x_i))$, assuming we know that $y' = \frac{\partial L}{\partial y}$, what is $x_i' = \frac{\partial L}{\partial x_i}$? **(8')**

Total 16

3. We will use the dataset below to learn a decision tree which predicts if people pass machine learning (Yes or No), based on their previous GPA (High, Medium, or Low) and whether or not they studied.

GPA	Studied	Passed
L	F	F
L	T	T
M	F	F
M	T	T
H	F	T
H	T	T

1) What is the entropy $H(Passed)$? **(4')**
2) What is the entropy $H(Passed \mid GPA)$? **(4')**
3) What is the entropy $H(Passed \mid Studied)$? **(4')**
4) Draw the best decision tree that would be learned for this dataset. Please state why the decision tree is the best. **(4')**

Total 16

4. Convolution operation is defined as follows:

$$f[x,y] * g[x,y] = \sum_{n_1=-\infty}^{\infty} \sum_{n_2=-\infty}^{\infty} f[n_1,n_2] \cdot g[x-n_1, y-n_2]$$

The image patch is defined as follows

$$f = \begin{pmatrix} 1 & 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 2 & 2 \\ 1 & 2 & 1 & 3 & 3 \\ 1 & 2 & 2 & 2 & 2 \\ 1 & 1 & 3 & 1 & 1 \end{pmatrix}$$

The convolution kernel is defined as

$$g = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

Please evaluate the convolution results for the 2 following cases:

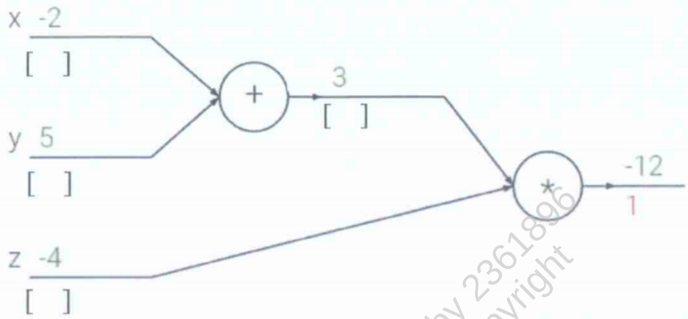
1) Stride is equal to 1, and **zero**-padding with 1 is used. **(10')**
2) Stride is equal to 2, and padding is **NOT** used. **(10')**

Total
20

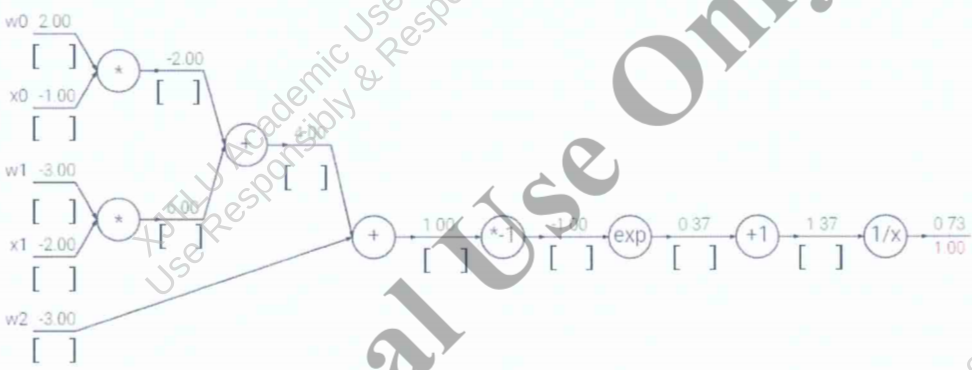
5. Fill in the missing gradients underneath the forward pass activations in each circuit diagram. The gradient of the output with respect to the loss is one (1.00), and has already been filled in. (The output values for each unit are represented on top of the line and the gradients are below the line.)

Please calculate the corresponding gradient **below** each unit in the brackets.

1)



2)



Total
16
(1 mark for each blank)

6. We will be working with clustering by GMM for data $\mathcal{D}$.

Assume a datapoint $\mathbf{x}$ is generated as follows:

- Choose a cluster z from $\{1, \ldots, K\}$ such that $p(z = k) = \pi_k$
- Given z, sample $\mathbf{x}$ from a Gaussian distribution $\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_z, \mathbf{I})$

1) What is the distribution of $p(x)$? **(5')** and What is the log-like loss function to be maximized? **(5')**
2) Illustrate the Expectation-Maximization algorithm for this GMM (Please list the steps for the EM algorithm) **(10')**

Total
20

The end of the paper